Kathmandu Valley Road Corridor Terrain and Drainage Analysis

This project analyses the terrain and drainage conditions along two major national highway corridors in Kathmandu Valley, NH34 Arniko Highway and NH41 Tribhuvan Highway. The study uses elevation data and GIS tools to understand how the landscape behaves around these roads, with a focus on slope conditions and natural drainage patterns.

Objective:

The project aims to identify slope zones along the highway corridors, map natural drainage channels and their catchment areas, and locate points where streams cross the road alignment, which are positions where culverts or bridges are required.

Methodology:

Road data for NH34 and NH41 was sourced from OpenStreetMap and filtered to include only the highway alignments within Kathmandu Valley. A 500 metre buffer was applied around both highways to define the study corridor. SRTM elevation data at 30 metre resolution was clipped to this corridor and used to derive slope, aspect, and contour layers. Watershed boundaries and stream networks were extracted using hydrological tools, and stream intersections with the road alignments were identified as drainage crossing points.

Outputs:

Three maps were produced. The first is a corridor overview showing both highway alignments within the valley. The second is a slope and contour map classifying terrain into flat, moderate and steep zones. The third is a drainage crossings map showing all 16 stream crossing locations along the two corridors. Supporting charts produced through Python show the elevation profile along the corridor and a comparison of terrain and drainage conditions between NH34 and NH41.

Key Findings:

The corridor spans an elevation range of 1167 metres to 1687 metres with a mean elevation of 1359 metres. Most of the corridor sits on relatively flat valley floor terrain, with steep zones concentrated near the valley rim. NH34 recorded 11 drainage crossings while NH41 recorded 5, reflecting the more active stream network on the eastern side of the valley.

Tools Used:

QGIS was used for all spatial analysis and map production. Python libraries including Rasterio, GeoPandas, NumPy and Matplotlib were used for terrain calculations and chart generation. Elevation data was sourced from SRTM via OpenTopography and road data from GeoFabrik.

Relevance:

The analysis reflects the kind of work carried out by geospatial engineering firms for infrastructure corridor surveys, road design support, and drainage planning. The outputs demonstrate practical application of GIS and Python in a civil engineering context.